File

Name: OPT3-2-K01
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1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1	---------- Gear 2 -------
Power (kW)	[P]	75.000	
Speed (1/min)	[n]	440.8	149.1
Number of load cycles (in mio.)	[NL]	528.960	178.913
Torque (Nm)	[T]	1624.8	4803.7
Application factor	[KA]	1.25	
Required service life (h)	[H]	20000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1	---------- Gear 2 -------
Center distance (mm)	[a]	274.207	
Center distance tolerance		ISO 286:2010 Measure js6	
Normal module (mm)	[mn]	6.0000	
Transverse module (mm)	[mt]	6.0000	
Normal diametral pitch (1/in)	[Pnd]	4.23333	
Transverse diametral pitch (1/in)	[Ptd]	4.23333	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	0.0000	
Number of teeth	[z]	23	68
Facewidth (mm)	[b]	28.60	28.60
Hand of gear		Spur gear	
Accuracy grade	[Q-ISO 1328:2013]	A6	A6
Inner diameter (mm)	[di]	68.04	329.50
Shaft diameter (mm)	[ds]	77.58	377.50
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1	---------- Gear 2 -------

3 Materials

Gear 1

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

Gear 2

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

		------- Gear 1	---------- Gear 2 -------
Surface hardness		HRC 61	HRC 61
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	425.00	425.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1500.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	850.00	850.00

3.1 Gear roughness

		------- Gear 1	---------- Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.60	0.60
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	4.80	4.80
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	Klüberoil GEM 1-220 N with details about wear coefficient kw	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	220.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	19.00
Specific density at 15°C (kg/dm³)	[ρ]	0.890
Oil temperature (°C)	[TS]	80.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	71.718
(mm)	[a0.e/i]	70.360 /70.292
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	2.377 / 2.382

Reference profile Gear 2

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000

Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	202.509
(mm)	[a0.e/i]	202.330 /202.248
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	2.505 / 2.510

4.1.1 Information on final machining

		------- Gear 1	---------- Gear 2 -------
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius reference profile	[ρfP*]	0.380	0.380
Addendum reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Type of profile modification:	none (only running-in)		
Tip relief by running in (µm)	[Ca L/R]	2.0 / 2.0	2.0 / 2.0

4.2 Basic data

Overall transmission ratio	[itot]	-2.957
Gear ratio	[u]	2.957
Transverse module (mm)	[mt]	6.000
Transverse pressure angle (°)	[αt]	20.000
Working pressure angle (°)	[αwt]	20.682
(°)	[αwt.e/i]	20.691 /20.673
Working pressure angle at normal section (°)	[αwn]	20.682
Base helix angle (°)	[βb]	0.000
Helix angle at operating pitch circle (°)	[βw]	0.000
Reference center distance (mm)	[ad]	273.000
Sum of profile shift coefficients	[Σxi]	0.2045

		------- Gear 1	---------- Gear 2 -------
Profile shift coefficient	[x]	0.4530	-0.2485
Generating profile shift coefficient	[xE.e/i]	0.4312/ 0.4198	-0.2783/-0.2920
Virtual gear no. of teeth	[zn]	23.000	68.000
Involute length (mm)	[l_dFa-l_dFf]	12.622	11.550

4.3 Diameters and their allowances

		------- Gear 1	---------- Gear 2 -------
Reference diameter (mm)	[d]	138.000	408.000
Base diameter (mm)	[db]	129.678	383.395
Tip alteration (mm)	[k*mn]	-0.020	-0.020
Tip diameter (mm)	[da]	155.396	416.978
(mm)	[da.e/i]	155.396 /155.296	416.978 /416.878
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.100	0.000 / -0.100
Tip form diameter (mm)	[dFa]	155.396	416.978
(mm)	[dFa.e/i]	155.396 /155.296	416.978 /416.878
Active tip diameter (mm)	[dNa]	155.396	416.978
(mm)	[dNa.e/i]	155.396 /155.296	416.978 /416.878
V-Circle diameter (mm)	[dv]	143.436	405.018
(mm)	[dv.e/i]	143.175 /143.038	404.661 /404.496

Operating pitch diameter (mm)	[dw]	138.610	409.804
(mm)	[dw.e/i]	138.618 /138.602	409.828 /409.780
Root diameter (mm)	[df]	128.436	390.018
(mm)	[df.e/i]	128.175 /128.038	389.661 /389.496
Active root diameter (mm)	[dNf]	133.044	398.333
(mm)	[dNf.e/i]	133.122 /133.024	398.407 /398.308
Root form diameter (mm)	[dFf]	132.668	395.168
(mm)	[dFf.e/i]	132.509 /132.427	394.916 /394.801

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	1.500	1.500
Effective tip clearance (mm)	[c.e/i]	1.827 / 1.663	1.765 / 1.615
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.347 / 0.258	1.803 / 1.696
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	8.698	4.489
(mm)	[ha.e/i]	8.698 / 8.648	4.489 / 4.439
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	4.782	8.991
(mm)	[hf.e/i]	4.913 / 4.981	9.170 / 9.252
Tooth height (mm)	[h]	13.480	13.480

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	37.832 /37.752	24.501 /24.463
Roll angle to dNf (°)	[ξNf.e/i]	13.292 /13.100	16.190 /16.136
Roll angle at dFf (°)	[ξFf.e/i]	12.038 /11.860	14.152 /14.080

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.9006	1.3899
Normal tooth thickness at tip circle (mm)	[san]	3.236	4.925
(mm)	[san.e/i]	3.193 / 3.072	4.834 / 4.731
Normal tooth thickness at tip form circle (mm)	[sFan]	3.236	4.925
(mm)	[sFan.e/i]	3.193 / 3.072	4.834 / 4.731
Normal space width at root circle (mm)	[efn]	0.000	5.062
(mm)	[efn.e/i]	0.000 / 0.000	5.116 / 5.142
Pitch on reference circle (mm)	[pt]	18.850	18.850
Base pitch (mm)	[pbt]	17.713	17.713
Transverse pitch on contact-path (mm)	[pet]	17.713	17.713

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	1.133	0.593
Specific sliding at the tip	[ζa]	0.573	0.464
Specific sliding at the root	[ζf]	-0.865	-1.343
Mean specific sliding	[ζm]	0.536	

Sliding factor on tip	[Kga]	0.354	0.186
Sliding factor on root	[Kgf]	-0.186	-0.354

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	28.600
Transverse contact ratio	[εα]	1.578
	[εα.e/m/i]	1.580 / 1.571/ 1.563
Overlap ratio	[εβ]	0.000
Total contact ratio	[εγ]	1.578
	[εγ.e/m/i]	1.580 / 1.571/ 1.563
Length of path of contact (mm)	[gα]	27.943
(mm)	[gα.e/m/i]	27.988 /27.834/27.679
Approach path of contact (mm)	[gf]	9.607
(mm)	[gf.e/m/i]	9.641 / 9.544/ 9.446
Recess path of contact (mm)	[ga]	18.335
(mm)	[ga.e/m/i]	18.347 /18.290/18.233

		Gear 1	Gear 2
Addendum contact ratio	[ε]	1.035	0.542
	[ε.e/i]	1.036 / 1.029	0.544 / 0.533
Length T1-A and T2-A (mm)	[T1A,T2A]	14.870	81.974
(mm)	[.e/i]	14.824 / 15.042	81.974 / 81.847
Length T1-B and T2-B (mm)	[T1B,T2B]	25.100	71.745
(mm)	[.e/i]	25.100 / 25.009	71.699 / 71.881
Length T1-C and T2-C (mm)	[T1C,T2C]	24.477	72.367
(mm)	[.e/i]	24.466 / 24.489	72.333 / 72.401
Length T1-D and T2-D (mm)	[T1D,T2D]	32.583	64.262
(mm)	[.e/i]	32.537 / 32.755	64.262 / 64.134
Length T1-E and T2-E (mm)	[T1E,T2E]	42.813	54.032
(mm)	[.e/i]	42.813 / 42.722	53.986 / 54.168
Length T1-T2 (mm)	[T1T2]	96.844	
(mm)	[.e/i]	96.799 / 96.890	
Diameter of single contact point B (mm)	[d-B]	139.055	409.366
(mm)	[d-B.e/i]	139.055 /138.989	409.334 /409.461
Diameter of single contact point D (mm)	[d-D]	145.130	404.363
(mm)	[d-D.e/i]	145.090 /145.285	404.363 /404.282

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	23547.35	
Axial force (N)	[Fa]	0.00	0.00
Radial force (N)	[Fr]	8570.53	8570.53
Normal force (N)	[Fnorm]	25058.56	25058.56
Nominal circumferential force per mm (N/mm)	[w]	823.33	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	23443.68	
Axial force (N)	[Faw]	0.00	0.00
Radial force (N)	[Frw]	8850.16	

Circumferential speed reference circle (m/s)	[v]	3.19	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	3.20	

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.8	
Running-in value (µm)	[yf]	0.9	
Correction factor	[CM]	0.800	
Gear blank factor	[CR, bs/b, sr/mn]	0.898	(0.250,5.000)
Basic rack factor	[CB]	0.975	
Material coefficient	[E/Est]	1.000	
Single stiffness (N/mm/µm)	[c']	12.695	
Theoretical single stiffness (N/mm/µm)	[c'th]	18.125	
Contact stiffness (N/mm/µm)	[cγα]	18.194	
Contact stiffness (N/mm/µm)	[cγβ]	15.465	
Reduced mass (kg/mm)	[mRed]	0.05720	
Resonance speed (min-1)	[nE1]	7405	
Resonance ratio (-)	[N]	0.060	
Subcritical range			

5.3 Calculation of K factors

Running-in value (µm)	[yα]	0.9	
Bearing distance l of pinion shaft (mm)	[l]	88.000	
Distance s of pinion shaft (mm)	[s]	8.800	
Outside diameter of pinion shaft (mm)	[dsh]	44.000	
Load in accordance with Figure 13, ISO 6336-1:2006	[-]	4	
0:a), 1:b), 2:c), 3:d), 4:e)			
Coefficient K' according to Figure 13, ISO 6336-1:2006	[K']	-0.60	
With stiffening			
Effective flank line deviation (µm)	[Fβy]	4.46	
from deformation of shaft (µm)	[fsh·B1]	2.03	
fsh (µm)	[fsh]	2.03	
B1	[B1]	1.00	
fHβ5 (µm)	[fHβ5]	7.00	
Tooth without flank line modification			
Position of contact pattern:	favorable		
from production tolerances (µm)	[fmα*B2]	13.09	
B_2=1.00			
Tooth trace deviation, theoretical (µm)	[Fβx]	5.24	
Running-in value (µm)	[yβ]	0.79	

5.4 K factors

Dynamic factor	[Kv]	1.019
Face load factors		
- Flank	[KHβ]	1.033
- Tooth root	[KFβ]	1.023
- Scuffing	[KBβ]	1.033
Transverse load factors		
- Flank	[KHα]	1.000
- Tooth root	[KFα]	1.000
- Scuffing	[KBα]	1.000
Application factor	[KA]	1.250
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.4198	-0.2920
Tooth form factor	[YF]	1.200	1.601
Stress correction factor	[YS]	2.239	1.839
Load application angle (°)	[αFen]	23.264	19.445
Load distribution influence factor	[fε]	1.000	
Load application diameter (mm)	[d_{en}]	145.130	409.366
Bending moment arm (mm)	[hFe]	5.894	7.338
(-)	[hFe*]	0.982	1.223
Tooth root thickness at critical cross section (mm)	[sFn]	13.147	12.870
(-)	[sFn*]	2.191	2.145
Root fillet radius at critical cross section (mm)	[ρF]	2.640	3.400
(-)	[ρF*]	0.440	0.567
Tangent contact point (mm)	[x,y]	6.574 /64.683	
Tangent contact point (mm)	[x,y]		6.435 /195.955
Diameter at critical cross section (mm)	[d_{sFn}]	130.033	392.120
Tangent at critical cross section (°)	[α_{sFn}]	30.000	30.000
Notch parameter	[q_s]	2.490	1.893
Helix angle factor	[Yβ]	1.000	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	28.60	28.60
Nominal stress at tooth root (N/mm²)	[σF0]	368.63	403.93
Tooth root stress (N/mm²)	[σF]	480.04	526.00
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	1.000	0.994
Surface factor	[YRrelT]	0.957	0.957
Size factor, tooth root	[YX]	0.990	0.990
Life factor	[YNT]	0.902	0.921
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.854	0.867
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)	[σFE]	850.00	850.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	518.41	526.63
Limit strength tooth root (N/mm²)	[σFG]	725.77	737.29

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	1.51	1.40
Transmittable power (kW)	[kWRating]	81.00	75.09

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.449	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.899	
Helix angle factor	[Zβ]	1.000	
Effective facewidth (mm)	[beff]	28.60	
Nominal contact stress (N/mm²)	[σH0]	1180.51	
Contact stress at operating pitch circle (N/mm²)	[σHw]	1353.86	

Single tooth contact factor	[ZB,ZD]	1.00	1.00
Contact stress (N/mm²)	[σHB, σHD]	1353.86	1353.86
Lubricant coefficient for N_L	[ZL]	1.020	1.020
Speed factor at N_L	[ZV]	0.973	0.973
Roughness factor for N_L	[ZR]	0.979	0.979
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.930	0.962
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.903	0.934
Limited pitting is permissible: No			
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	1354.63	1400.43
Pitting stress limit (N/mm²)	[σHG]	1354.63	1400.43

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	1.00	1.03
Safety against pressure, σHG/σHBD Single contact	[SHBD]	1.00	1.03
Safety regarding transmittable torque	[SHBD²]	1.00	1.07
Transmittable power (kW)	[kWRating]	75.09	80.25

8 Micropitting

Calculation method according to ISO/TS 6336-22:2018
Lubricant load according to FVA Info sheet 54/7 10, Klüberoil GEM 1-220 N with details about wear coefficient kw
Reference data FZG-C Test:

Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	127.666
Contact temperature (°C)	[θBRef,A]	251.735
Lubrication gap thickness (µm)	[hRef,A]	0.051
Specific film thickness in test	[λGFT]	0.102

		------- Gear 1	---------- Gear 2 -------
Calculation of permissible specific film thickness			
Material coefficient	[WW]	1.000	
Permissible specific film thickness	[λGFP]	0.143	
Interim results in accordance with ISO/TS 6336-22:2018			
Coefficient of friction	[μ_m]	0.069	
Lubricant factor	[XL]	1.000	
Roughness factor	[XR]	0.936	
Lubrication coefficient for lubrication type	[XS]	1.000	
Tooth mass temperature (°C)	[θM]	87.851	
Tip relief factor	[XCa(A)]	1.000	
Loss factor	[HV]	0.144	
Equivalent Young's modulus (N/mm²)	[Er]	226373.626	
Pressure-viscosity coefficient (m²/N)	[α38]	0.02162	
Dynamic viscosity (Ns/m²)	[ηtM]	22.951	
Roughness average value (µm)	[Ra]	0.600	

Calculation of speeds, load distribution and flank curvature according to method B following ISO/TS 6336-22:2018.

C_a taken as optimal in the calculation. 0=no, 1=yes	0	0
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)	0	

Diameter (mm)	[dy]	133.044	416.978
Relative radius of curvature (mm)	[ρred]	12.587	
Load sharing factor	[XY]	0.333	
Contact stress (N/mm²)	[pH]	914.330	
Contact stress (N/mm²)	[pdyn]	1048.587	
Minimal specific film thickness	[λGFY]	0.203	(h_Y=0.122 µm)
Safety against micropitting	[SΛ(B)]	1.419	

9 Safety of hardened layer (subsurface fatigue)

CHD according to	DNV-CG-0036:2021		
		------- Gear 1 ----------	Gear 2 -------
Required hardening depth CHD (mm)	[t550]	0.50 / 0.60	0.50 / 0.60
Required hardness at t550 (HV)	[HV.t]	550	550
Required hardening depth CHD (mm)	[t400]	0.90 / 1.10	0.90 / 1.10
Required hardness at t400 (HV)	[HV.t]	400	400
Required hardening depth CHD (mm)	[t300]	1.30 / 1.60	1.30 / 1.60
Required hardness at t300 (HV)	[HV.t]	300	300
Condition for hardness on the surface (HV)	[> HV.0]	599	599
Condition for hardness on the surface (HRC)	[> HRC.0]	55	55

HV.0 is calculated with t_z/a_H = 0.5, S_H = min(S_{EHT}, S_{Hssmin})

Safety of the hardened layer	[SEHT]	0.99	0.99
Required safety	[SHssmin]	1.00	1.00
Result		2	2

-1:Not checked, 0:CHD too shallow, 1:CHD too deep or HV.0 too large, 2:Check performed,
3:Check performed with insufficient hardening depths (t_z/a_H<0.5, t_z/a_{Hst}<0.6)

10 Tooth flank fracture

Calculation method according to ISO/TS 6336-4 (2019)

Calculation according to Method B

		------- Gear 1 ----------	Gear 2 -------
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)		2	2
Diameter (mm)	[dy]	139.055	409.366
Relative radius of curvature (mm)	[ρred]	18.594	18.594
Load sharing factor	[XCP]	1.000	1.000
Contact stress (N/mm²)	[pH]	1302.945	1302.945
Contact stress (N/mm²)	[pdyn]	1494.266	1494.266
Required hardening depth CHD (mm)	[t550]	0.50 / 0.60	
Required hardening depth CHD (mm)	[t550]		0.50 / 0.60
The calculation will be performed with the minimum hardening depth			
Core hardness (HV)	[HV_{core}]	342	342
Hardness curve from:			
HV(y) gear 1		ISO/TS 6336-4, chapter 7.3.3, Method C1	
HV(y) gear 2		ISO/TS 6336-4, chapter 7.3.3, Method C1	
Maximum material exposure	[AFFmax]	1.264	1.264
Depth of maximum load (mm)	[y]	-1.050	-1.050
Material coefficient	[$K_{material}$]	1.131	1.131
Material shearing strength (N/mm²)	[τper]	164.193	164.250
Equivalent stress state (N/mm²)	[τeff]	200.986	200.986
Safety factor according to ISO DTS 19042-1	[SFF]	0.883	0.883

(S_{FF} = 1/($A_{FF,max}$ + 0.2) + 0.2)

11 Scuffing load capacity

Calculation method according to		ISO/TS 6336-20/21:2022		
		------- Gear 1	---------- Gear 2	-------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780	
Relevant tip relief (µm)	[Ca]	2.00	2.00	
Optimal tip relief (µm)	[Ceff]	81.07		
C_a taken as optimal in the calculation. 0=no, 1=yes		0	0	
		---------- Pair	----------	
Helical load factor for scuffing	[KBγ]	1.000		
Lubrication coefficient for lubrication type	[XS]	1.000		
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	14	
Multiple meshing factor	[Xmp]	1.000		
Relative structural factor, scuffing	[XWrelT]	1.000		
Effective facewidth (mm)	[beff]	28.600		
Applicable circumferential force/facewidth (N/mm)	[wBt]	1082.877		
$K_{B\gamma}$ = 1.000, $w_{Bt} \cdot K_{B\gamma}$ = 1082.877				
Angle factor	[Xαβ]	0.988		
ε_1 =1.035, ε_2 =0.542				

11.1 Flash temperature-criteria

		---------- Pair	----------	
Lubricant factor	[XL]	0.844		
Tooth mass temperature (°C)	[θMi]	94.01		
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$				
Average flash temperature (Δ°C)	[θflm]	29.81		
Scuffing temperature (°C)	[θS]	451.99		
Γ coordinates (point of highest temperature)	[Γ]	0.645		
[Γ.A]= -0.393, [Γ.E]= 0.749				
Maximum flash temperature (Δ°C)	[θflamax]	48.08		
Maximum contact temperature (°C)	[θBmax]	142.09		
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058		
Approach factor	[XJ]	1.000		
Load sharing factor	[XΓ]	0.416		
Dynamic viscosity (mPa*s)	[ηM]	30.07	(80.0	°C)
Coefficient of friction	[μ_m]	0.075		
Required safety	[SBmin]	2.000		
Margin of safety for scuffing, flash temperature	[SB]	5.991		

11.2 Integral temperature-criteria

		---------- Pair	----------	
Lubricant factor	[XL]	1.000		
Tooth mass temperature (°C)	[θMC]	94.32		
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$				
Integral scuffing temperature (°C)	[θSint]	455.17		
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058		
Running-in factor, well run in	[XE]	1.000		
Contact ratio factor	[Xε]	0.244		
Dynamic viscosity (mPa*s)	[ηOil]	30.07	(80.0	°C)
Mean coefficient of friction	[μ_m]	0.068		
Geometry factor	[XBE]	0.332		
Meshing factor	[XQ]	1.000		
Tip relief factor	[XCa]	1.106		
Mean flash temperature (Δ°C)	[θflaint]	20.46		
Integral tooth flank temperature (°C)	[θint]	125.01		

Required safety	[SSmin]	1.800
Safety factor for scuffing (intg.-temp.)	[SSint]	3.641
Safety factor for transmitted moment (int.-T.)	[SSL]	8.336

12 Measurements for tooth thickness

12.1 Tooth thickness tolerances

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.095 / -0.145	-0.130 / -0.190

12.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
Number of teeth spanned	[k]	4.000	8.000
Base tangent length (no backlash) (mm)	[Wk]	65.787	137.540
Base tangent length with allowance (mm)	[Wk.e/i]	65.697 /65.650	137.418 /137.362
(mm)	[ΔWk.e/i]	-0.089 / -0.136	-0.122 / -0.179
Diameter of measuring circle (mm)	[dMWk.m]	145.359	407.268

12.3 Measurement over balls and pins

		------- Gear 1 ----------	Gear 2 -------
Theoretical diameter of ball/pin (mm)	[DM]	11.537	9.996
Effective diameter of ball/pin (mm)	[DMeff]	12.000	10.000
Radial single-ball measurement, no backlash (mm)	[MrK]	80.835	209.180
Radial single-ball measurement (mm)	[MrK.e/i]	80.745 /80.698	209.002 /208.920
Diameter of measuring circle (mm)	[dMMr.m]	143.856	404.616
Diametral measurement over two balls, no backlash (mm)	[MdK]	161.320	418.360
Diametral measurement over two balls (mm)	[MdK.e/i]	161.142 /161.047	418.005 /417.839
Diametral measurement over pins, no backlash (mm)	[MdR]	161.320	418.360
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	161.142 /161.047	418.005 /417.839
Measurement over 3 pins, no backlash (mm)	[Md3R]	160.972	0.000
Measurement over 3 pins (mm)	[Md3R.e/i]	160.794 /160.700	0.000 / 0.000

12.4 Tooth thickness

		------- Gear 1 ----------	Gear 2 -------
Medium tip diameter (mm)	[da.m]	155.346	416.928
Reference chordal height from da.m (mm)	[hac]	8.909	4.507
Tooth thickness at height hac, chord (mm)	[sc]	11.390	8.339
(mm)	[sc.e/i]	11.298 /11.249	8.210 / 8.150
Tooth thickness on reference circle, arc (mm)	[sn]	11.403	8.339
(mm)	[sn.e/i]	11.308 /11.258	8.209 / 8.149

12.5 Backlash

		---------- Pair ----------
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.012 / -0.012
Radial backlash (mm)	[jrw.e/i]	0.464 / 0.284
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.349 / 0.214
Normal backlash (mm)	[jnw.e/i]	0.326 / 0.200
Center distance allowances (mm)	[Aa.e/i]	0.016 / -0.016
Backlash free center distance (mm)	[aControl.e/i]	273.907 /273.759

Backlash free center distance, allowances (mm)	[jta]	-0.300 / -0.448	

		------- Gear 1 ----------	Gear 2 -------
dNf.i with aControl (mm)	[dNf0.i]	132.497	397.648
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	-0.006	1.366
Tip clearance (mm)	[c0.i(aControl)]	1.230	1.182

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.0975/ 0.0598	(0°5'51"/ 0°3'35")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.2882/ 0.1768	(0°17'17"/ 0°10'37")

13 Gear tolerances

		------- Gear 1 ----------	Gear 2 -------
According to ISO 1328-1:2013, ISO 1328-2:1997			
Accuracy grade	[Q]	A6	A6
Single pitch tolerance (µm)	[fpT]	±11.00	±11.00
Base pitch tolerance (µm)	[fpbT]	±10.34	±10.34
Sector pitch tolerance (µm)	[Fpz/8T]	±22.00	±26.00
Cumulative pitch tolerance, total (µm)	[FpT]	32.00	40.00
Adjacent pitch difference tolerance (µm)	[fuT]	15.00	16.00
Profile form tolerance (µm)	[ffαT]	12.00	12.00
Profile slope tolerance (µm)	[fHαT]	±9.00	±9.50
Profile tolerance, total (µm)	[FαT]	15.00	15.00
Helix form tolerance (µm)	[ffβT]	10.00	11.00
Helix slope tolerance (µm)	[fHβT]	±9.00	±9.50
Helix tolerance, total (µm)	[FβT]	14.00	15.00
Runout tolerance (µm)	[FrT]	29.00	36.00
Single flank composite tolerance, total (µm)	[FisT]	43.00	50.00
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	10.00	10.00
Radial composite tolerance, total (µm)	[Fi"]	51.00	60.00
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	22.00	22.00

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020			
Accuracy grade	[R]	R41	R41
Radial composite tolerance, total (µm)	[FidT]	45.00	57.00
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	21.00	24.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99

Total helix deviation for calculation (µm)	[Fβ]	15.00
Distance between bearings for calculation (mm)	[LG]	88.00
Maximum value for axis deviation error (µm)	[fΣβ]	± 23.08
Maximum value for axis inclination error (µm)	[fΣδ]	± 46.15

14 Modifications and determination of the tooth form

14.1 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

15 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	277.985

15.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (kg)	[m]	2.728	9.540
Total mass (kg)	[Σm]	12.268	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.00845	0.32363
System (da+df)/2...di (kg*m²)	[J]	0.04547	

15.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	1.851
Torsion when subjected to nominal torque (°)	[δcr]	0.050
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	78.319

15.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.069
Wear sliding coefficient by Niemann	[ζw]	0.845
Loss factor	[HV]	0.144
Gear power loss (kW)	[PVZ]	0.740
Meshing efficiency (%)	[ηz]	99.013

16 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	20000.0000
Service life (calculated with required safeties):		
System service life (h)	[H_{att}]	20381

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	2.124e+04
Tooth flank service life (h)	[H_{Hatt}]	2.038e+04	6.026e+04

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

16.1 Damage

Damage relative to the required service life (H, 20000.0 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	94.1433	98.1310	33.1914

Damage relative to the system service life (H_{att}, 20380.9 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	95.9363	100.0000	33.8235

17 Reliability calculation

17.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

17.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	β	t_0	T	R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	26448	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	26448	1.300e+00	4.859e+08	2.315e+09	99.2382
2	Tooth root	8946	1.700e+00	1.835e+08	2.819e+08	100.0000
2	Tooth flank	8946	1.300e+00	4.859e+08	2.315e+09	100.0000

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

17.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	20000.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	99.2382
Reliability R, gears subsystem (%)	[R_{subG}]	99.2382
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	21244.2173
Service life H, tooth flanks subsystem (h)	[H_{subH}]	20380.9186
Service life H, gears subsystem (h)	[H_{subG}]	20380.9185

18 Remarks

18.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

18.2 Calculations and factors

- Details of calculation method:
  c_v according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 733)